



USN

Nitte (Deemed to be University)

(Established under Sec. (3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CST131 – REPAIR AND RESTORATION OF STRUCTURES

(For CST)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit-I** and **Unit-II** and **One full** question from **Unit-III**.

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) Explain with a neat sketch the working principle and procedure rebound of hammer test. State its objectives.	10	L2	1	1
	b) What is corrosion mapping? Explain any two methods adopted for mapping of corrosion.	10	L2	2	1
2.	a) What are the causes of deterioration of concrete? Explain any three.	10	L2	1	1
	b) Discuss any four methods adopted for reducing initiation of corrosion.	10	L2	2	2
3.	a) Discuss Ultrasonic Pulse Velocity (UPV) Test and Ground Penetrating Radar (GPR), with respect to working principle, applications and limitations.	10	L2	1	2
	b) Explain the strength and thermal properties of built-up concrete.	10	L2	2	2
		Unit – II			
4.	a) Discuss the necessity of maintenance with respect to atmospheric agencies.	10	L2	3	1
	b) Discuss the applications of foamed concrete in repair of concrete structures.	10	L2	4	1
5.	a) List the importance of maintenance and explain any two.	10	L2	3	1
	b) Discuss the process of rust elimination by polymer coating.	10	L2	4	1
6.	a) Enumerate the classification of maintenance and discuss any two.	10	L2	3	1
	b) Briefly explain shoring and underpinning, with neat sketches.	10	L2	4	1
		Unit – III			
7.	Explain any four methods adopted to strengthen the low-strength flexural members.	20	L2	5	1
8.	Explain any two methods adopted to strengthen the foundation of a structure.	20	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
